

# SK Saif Ibna Ezhar Arko

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## Research Interests

Quantum Information Theory • Quantum Computing and Algorithms • Quantum Cryptography • Quantum Machine Learning • Adversarial Robustness of Quantum Models • Post-Quantum Cryptography.

## Education

**BRAC University**, Dhaka, Bangladesh

2026

*B.Sc. in Computer Science and Engineering* — CGPA: **3.20 / 4.00** (Approx. 78.33% equivalent based on grade-boundary interpolation)

- **Undergraduate Thesis:** *Quantum-Aware Image Encoding and Adversarial Perturbation*
- **Relevant Coursework:** Quantum Computing, Quantum Error Correction, Quantum Algorithms, Machine Learning, Pattern Recognition, Data Structures & Algorithms, Linear Algebra, Probability & Statistics.

**Notre Dame College**, Dhaka

2020

*Higher Secondary Certificate (Science)* — GPA: **5.00 / 5.00**

**St. Gregory's High School and College**, Dhaka

2018

*Secondary School Certificate (Science)* — GPA: **5.00 / 5.00**

## Research Experience

**Undergraduate Thesis Researcher** — BRAC University, Dept. of CSE

2024 – Present

*Supervisor: Ipshita Bonhi Upoma, Lecturer, Department of CSE, BRAC University*

- Developing theoretical frameworks for encoding classical image data into quantum states via amplitude and basis encoding, and analyzing their robustness against adversarial noise.
- Studying trade-offs between encoding fidelity, circuit depth, and resilience to bit-flip and phase-flip channels.

**Student Researcher** — ELITE Research Lab LLC

Sept. 2026 – Jan. 2027

*Supervisor: Md Kishor Morol*

- Conducting research at the intersection of quantum computing and machine learning in collaboration with industry experts.

**Research Intern, QIntern 2026 Global Internship** — QWorld

Summer 2026

*Supervisor: Dr. Maha A. Metawei*

- Project: *Accelerating hybrid quantum-classical machine learning tasks on HPC platforms (QNLP)*.
- Implemented high-performance quantum circuit simulations and tensor network optimizations to scale hybrid algorithms on distributed clusters; extended core framework classes (*Ansatz*, *Reader*) and built unified data format conversion scripts.

**Research Intern, QIntern 2025 Global Internship** — QWorld

Summer 2025

*Supervisors: Dr. Kumar Prateek, Dr. Soumyadev Maity*

- Project: *Design of Quantum-Anamorphic Encryption Against Coercive Surveillance*. Prototyped anamorphic quantum encryption schemes providing deniability against coercive adversaries.
- **Awarded 3rd Best Project globally** among submissions worldwide; collaborated with an international team across multiple countries.

## Professional Experience

**Datasoft Systems Bangladesh Limited**, Dhaka

December 2025 – Present

*Software Developer (AI and Quantum Department)*

- Deploy and operate self-hosted LLM inference infrastructure with `llama.cpp` and `llama-server`, serving a 228B-parameter (10B active) Mixture-of-Experts model in quantized GGUF format across heterogeneous single-GPU nodes.
- Designed the *Hermes Agent* framework: model-switching, multi-provider LLM routing (NVIDIA NIM, DeepSeek, OpenRouter, local Ollama) behind an authenticated API gateway running as a supervised `systemd` service.
- Built LangGraph-based agentic SQA workflows executing multi-step verification across Django/React/PostgreSQL projects, and delivered *iSAGE*, an internal assistant UI over the local LLM backend.
- Engineer production Python systems for document processing and workflow automation, including distributed scraping and ETL pipelines; contributed to the terminal operating system of Chittagong Sea Port.

- Engineered Python tools for web scraping, automation, database management, and REST API integration.

## Selected Projects

### QVTM — Quantum Variable Topic Model (Global Quantum Hackathon 2024) [\[GitHub\]](#)

- Built an encoder–decoder variational quantum model using **amplitude embedding** to represent text in Hilbert space, optimized with a composite **KL-divergence**, **Cross-Entropy**, and **MMD** objective.
- Achieved a **Topic Diversity score of 1.0** with improved coherence over Latent Dirichlet Allocation baselines.

### Monkeypox Classification Using a Hybrid Quantum CNN (HQCNN) [\[GitHub\]](#)

- Combined classical feature extraction with parameterized quantum circuits and benchmarked against purely classical CNN and EfficientNetB3 transfer-learning baselines for medical image diagnosis.

### Lexora — Bengali-First English Vocabulary Learning Platform [\[Live\]](#) [\[GitHub\]](#)

- Built a Django 5 and HTMX progressive web app (~9,900 lines of Python across five apps, 89 tests) for GRE and IELTS preparation, serving 1,468 words in 62 sequential packs with paired Bangla and English meanings.
- Implemented a full **SuperMemo-2** spaced-repetition scheduler alongside a Leitner five-box ladder and an **Ebbinghaus retention-curve** exam forecast, plus study modes spanning timed drills, dictation, confusable-pair training, and an AI-driven debate arena.

### NLP Similarity Classifier — Bidirectional LSTM Siamese Network [\[GitHub\]](#)

- Designed a TensorFlow/Keras Siamese network for paraphrase identification, fusing sentence embeddings with Jaccard and fuzzy-matching features across four merge strategies; an augmentation pipeline generated **7,400+ synthetic samples**, evaluated via ROC curves and confusion matrices.

**Other projects:** [Stock price forecasting — ARIMA vs. LSTM](#); [SmartCare](#), a Django healthcare management platform; [3D Ball Roller](#), a physics-based OpenGL game in Python.

## Community Contributions, Teaching, and Professional Service

### IBM Qiskit Advocate (#129943) — IBM Quantum 2025 – Present

- Recognized for sustained contributions to the global quantum community; lead workshops, mentor early-career researchers on Qiskit projects, create tutorials and learning resources, support the community on official Discord and GitHub channels, and organize outreach events and hackathons.

### QBangladesh — National Hub of QWorld, Bangladesh April 2025 – Present *Founder, Coordinator, and Mentor*

- Established and lead the first national quantum computing community in Bangladesh under the QWorld umbrella, building a sustainable ecosystem for quantum education and research.
- Deliver hands-on workshops on quantum computing with Python, Qiskit, and Cirq for students nationwide, and mentor early-career researchers on quantum algorithms and simulation.

### Caretutors Technology Limited, Dhaka — *Verified Tutor* December 2024 – Present

- Deliver Python instruction covering programming logic, data structures, and core language concepts through personalized, project-based learning plans.

## Technical Skills

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|-------------------------|------------------------------------------------------------------------|
| <b>Programming:</b>     | Python, C, C++, SQL                                                    |
| <b>Quantum:</b>         | Qiskit, Cirq, PennyLane (basic), Stim                                  |
| <b>ML / DL:</b>         | TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, LangGraph  |
| <b>Web &amp; Tools:</b> | Django, HTMX, PostgreSQL, HTML/CSS, OpenGL, Git, Linux, Jupyter, LaTeX |
| <b>Languages:</b>       | Bengali (Native), English (Full Professional Proficiency)              |

## Honors, Awards, and Certifications

**Awards:** Best Project (2nd Place), “Accelerating Hybrid Quantum-Classical Machine Learning Tasks on HPC Platforms,” QIntern 2026 (QWorld, September 2026); Outstanding Participation Diploma, QIntern 2026 (QWorld); 3rd Best Project Globally, QIntern 2025 (QWorld); Diploma in Quantum Anamorphic Encryption, QWorld 2025.

**Certifications:** ICTP Physics Without Frontiers: Bangladesh School on Mathematics for Physics 1 — Operators on Hilbert Spaces (ICTP, 2026); Quantum AI Lab — Quantum Cryptography (Bibliotheca Alexandrina, 2026); Quantum Business Foundations (IBM, 2026); Basics of Quantum Information (IBM, 2026); Pan-African Quantum Challenge 2025 (Africa Quantum Consortium, 2026); Quantum Random Number Generation (Quantum Security Defence, 2026); Introduction to Quantum Computing (Qubit by Qubit, 2025); QCOURSE 501-2: Elements of Quantum Computing and Programming (QWorld, 2025); QSilver26 and QBronze144 Workshops (QWorld, 2024); Quantum Machine Learning Fundamentals (Ingenii, 2024).

## References

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### **Ipshita Bonhi Upoma**

Lecturer, Dept. of CSE, School of Data and Sciences  
BRAC University, Bangladesh (*Thesis Supervisor*)  
[ipshita.upoma@bracu.ac.bd](mailto:ipshita.upoma@bracu.ac.bd)

### **Dr. Kumar Prateek**

Assistant Professor (Grade-II)  
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